

LIN HE

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PROFILE

AI/ML Engineer with products experience shipping ML systems in a regulated cybersecurity environment. Combines classical Machine Learning algorithms and Deep Learning foundations with full-stack delivery — from model adaptation and training, user-oriented front-end to inference backend, cloud infrastructure, and production monitoring. Interested in SOTA AI and LLM development, agent systems, and inference optimization roles to deliver real production level solutions.

PROJECTS

Enterprise AI Orchestration Platform (Automated SOC Agent)

[Github/Lin0He](#)

LangGraph · FastAPI · pgvector · React · Docker

- **Stateful LLM Orchestration:** Architected an event-driven agent engine using LangGraph, enforcing deterministic execution over non-deterministic LLM outputs; modelled incident-triage workflows as typed state graphs with conditional branching, retry semantics, and structured output enforcement via Pydantic + instructor — eliminating JSON parse failures in production.
- **Secure MCP Gateway:** Engineered a Model Context Protocol gateway from scratch with custom JWT-based authorization, creating a strict privilege boundary between the LLM reasoning layer and privileged tool execution (host isolation, firewall rules) — zero unauthorized tool calls in adversarial testing.
- **Hybrid RAG Pipeline:** Built an end-to-end RAG system combining pgvector semantic search with BM25 keyword retrieval and RRF re-ranking over cybersecurity playbooks; dynamically injects ranked context into the LLM prompt, improving retrieval precision for low-frequency domain-specific terms.
- **LLM Evaluation & Observability:** Implemented a RAG evaluation pipeline using RAGAS metrics (faithfulness, context recall, answer relevance) to monitor retrieval quality across prompt iterations; integrated LangSmith distributed tracing for full observability of agent reasoning chains, token usage, and tool-call success rates per workflow step.
- **Human-in-the-Loop Architecture:** Designed conditional graph edges to automatically suspend high-risk execution paths and await explicit analyst approval via a React operator console, eliminating hallucination risk for destructive actions in production.

LLM Quantization and Inference Optimization Benchmarking

[Github/Lin0He](#)

PyTorch · CUDA · vLLM · Llama-3 · Mistral-7B

- **Quantization & Accuracy Fidelity:** Applied 4-bit quantization (GPTQ/NF4) to Llama-3 and Mistral-7B, achieving a 70% VRAM reduction (15.5 GB → 5 GB) with < 2% accuracy drift on GSM8K and HumanEval — enabling single-GPU deployment without quality loss.
- **Inference Profiling Suite:** Built a custom Python profiling framework to instrument TTFT and TPS across prefill and decode stages; identified memory-bandwidth as the primary bottleneck and applied operator fusion and CUDA kernel optimizations to maximize TFLOPS utilization on RTX 4090.
- **KV Cache & Architecture Analysis:** Analysed KV cache scaling and memory-bandwidth tradeoffs between GQA (Mistral) and MHA architectures; established a Pareto-optimal configuration space (batch size × context length) to guide deployment decisions for high-concurrency local inference.

EXPERIENCE

Research Engineer, Machine Learning / AI

Feb 2025 – Present

Irdeto

Hoofddorp, NL

Selected for the **Engineer Graduate Program** to independently own end-to-end delivery of AI/ML initiatives across Irdeto's global cybersecurity and anti-piracy division — from idea PoC (Proof of Concept) and prototypes to production deployment and scale-up.

- **Domain-Adaptive LoRA Fine-tuning & evaluation:** Fine-tuned a domain-specific LLM using LoRA (PEFT) on company cloud infrastructure for cybersecurity log analysis; designed training data curation pipeline, evaluation framework, and iterative prompt-alignment workflow — adapting general-purpose base models to production-grade domain knowledge requirements.

- **End-to-End ML System Ownership:** Owned the full ML lifecycle of a Server Load Performance Prediction Service (time-series forecasting) deployed across EMEA and APAC — feature engineering, model selection, training, evaluation, and production deployment on AWS; achieved > 80% prediction accuracy and delivered stakeholder documentation and customer support content for production rollout.
- **AI Model Benchmarking & Evaluation:** Designed an adversarial robustness evaluation framework for an AI-based watermark detection system; benchmarked model variants under real-world perturbations (compression, cropping, noise injection) across precision/recall tradeoffs — establishing reproducible evaluation methodology for production model selection.
- **MLOps & Inference Infrastructure:** Implemented a high-performance inference serving backend (REST API, Docker) integrated with the internal security platform; automated ML data pipelines and deployment on AWS (S3, ECS, Lambda, SageMaker, CloudWatch) with CI/CD monitoring; integrated React front-end dashboard for real-time model output visualization.

Machine Learning Engineer Intern
supplai(by visionplatform.ai)

March 2023 – Dec 2023
Rotterdam, NL

- **Edge AI Inference Pipeline:** Built an end-to-end real-time inference pipeline from scratch for GigE industrial camera input on NVIDIA Jetson embedded hardware; deployed and optimized SOTA OCR models achieving > 95% accuracy in production, handling live video stream preprocessing, model inference, and automated output processing.

EDUCATION

Leiden University
Master of Science in **Computer Science** (Artificial Intelligence Track)
Thesis: [Classification of Vocal Intensity Categories using ResNet](#) — Applied ResNet with multi-modal audio feature extraction (Spectrogram, Mel-spectrogram, MFCC) to classify vocal intensity levels; achieved 71% accuracy on the Aalto Vocal Intensity Dataset without calibration signal dependency.
Relevant coursework: Deep Learning, Computer Vision, Probabilistic ML, Optimization, NLP, Acoustic & Multimedia Systems

Sep 2022 – Jan 2025
GPA: 8/10

Jiangsu University
Bachelor of Engineering in **Automation**
Control Engineering combines Applied Mathematics, Electrical Engineering, Embedded System, Robotics, etc. Strong manufacturing backgrounds in automotive electronics and embedded system control.

Sep 2018 – Jun 2022
GPA: 86/100

SKILLS

Programming	Python, C/C++, JavaScript, SQL, Shell/Bash
LLM & Agents	LangChain, LangGraph, HuggingFace, PEFT/LoRA, DPO, vLLM
ML Systems	PyTorch, CUDA, MLflow, Polars, Scikit-learn, Weight and Bias, ChromaDB
Infrastructure	AWS (SageMaker, ECS, Lambda, S3, DynamoDB, CloudWatch, etc.), Docker, Terraform, CI/CD, Git
Focus Areas	GenAI/LLM Engineering & Agent Systems, Inference Optimization, AI System Evaluation, Deep Learning
Languages	Chinese (Native), English & Japanese (Full Professional), Dutch & French(Beginner)